Declaration of Academic Integrity

Academic Integrity is a central tenet of Singapore Polytechnic. The polytechnic rules state that "Cheating in examinations and other assessed work is a very serious offence. This includes copying and using plagiarised material. Any student who cheats, attempts to cheat or breaches any rules for examinations and tests will face disciplinary action. The student is liable to be expelled."

Check **only one** of the two options below:

☐ I affirm that the work I submit is my own, produced without help from any AI tool(s) and/or other source(s).

☑ I affirm that the work I submit has been produced with the use of AI tool(s) and/or other source(s) which I have acknowledged fully in the following section.

By signing this form, I declare that the above affirmation made is true, and that I have read and understood the rules stated in Students Handbook on "Plagiarism" and "Breach of Examination/Assessment Rules".

Name: Ethan Chuwa Si Xuan	Student ID: P2508638	Class: DAAA/FT/1B/04
Signature: X	Date: 30/11/2025	

Acknowledgement of use of AI tool(s) and/or other source(s) (where applicable)

AI tool(s)

Name of AI tool	ChatGPT
Input prompt	Explain the difference between cross_val_score() and cross_val_predict() and advise which should be used for model evaluation in my regression task.
Date generated	10 Nov 2025
Output generated	cross_val_score computes aggregated metric scores for model performance, while cross_val_predict returns out-of-fold predictions for building plots such as Actual vs Predicted and Residuals vs Predicted.
Impact on submission	Helped me understand the situations in where each is used and allowed me to correctly use the code.

Name of AI tool	ChatGPT
Input prompt	Show me an example of how to use cross_val_predict
Date generated	15 Nov 2025

Output generated	kf = KFold(n_splits=3, shuffle=True, random_state=42)
	
	# ----------
	# cross_val_predict → gives OUT-OF-FOLD predictions
	# ----------
	y_pred_log = cross_val_predict(
	  pipe,
	  X,
	  y,
	  cv=kf
	)

Impact on submission	Gives me an idea on how to apply cross_val_predict

Other source(s)

Source	**Impact on submission**
< Link to online source >	*< Briefly explain which part of your submitted work the online source was applied >*
< Author, Book / Article / Journal, Year, Pages >	*< Briefly explain which part of your submitted work the Book / Article / Journal was applied >*
< Student / Person name >	*< Briefly explain which part of your submitted work the help of another student / person was applied >*
< Others >	*< Briefly explain which part of your submitted work the source was applied >*